Khwaja Yunus Ali University

School of Science and Engineering
Department of Computer Science and Engineering
Mid-term Examination: Spring Semester 2021
Program: B.Sc. in CSE (Batch – 12^{th}) 1^{st} Year 1^{st} Semester
Course Title: Physics - I
Course Code: PHY 1101

Time: 10 Minutes **Total Marks: 10 X 1 = 10**

Student ID:	**Batch:**	**Session:**

Part 1: MCQ

These are single best answer type of questions. Each of the following question or statement has five suggested answers or completions. Write down the most appropriate one on your answer script.

1. Which is the unit of current density?

   a) $Volt$
   b) Am^{-1}
   c) Am^{-2}
   d) JC^{-1}
   e) CV^{-1}

2. The potential differences between two points in an equipotential surface is ……………

   a) 0
   b) 1
   c) 2
   d) 3
   e) 4

3. The reciprocal of resistance is called the –

   a) resistivity
   b) specific resistance
   c) conductance
   d) conductivity
   e) permittivity

4. With an equality or balance of charge, the object said to be-

   a) maximum charged
   b) negatively charged
   c) positively charged
   d) electrically neutral
   e) mixed charged

5. What is the value of the minimum amount of charge in universe?

   a) $1.6 \times 10^{-16}C$
   b) $1.6 \times 10^{-19}C$
   c) $9.1 \times 10^{-31}C$
   d) $1.9 \times 10^{-16}C$
   e) $3.1 \times 10^{-34}C$

6. Which one is the right equation?

   a) $V = -\nabla E$
   b) $\bar{E} = -\bar{\nabla}\mathrm{V}$
   c) $\bar{F} = -\bar{\nabla}\mathrm{V}$
   d) $F = -\nabla V$
   e) $E = -\nabla R$

P.T.O.

7. What is the permittivity of the medium of dielectric constant 1.5?

a) $1.5 \times 10^{-12} Fm^{-1}$

b) $8.85 \times 10^{-12} Fm^{-1}$

c) $13.3 \times 10^{-12} Fm^{-1}$

d) $15.5 \times 10^{-12} Fm^{-1}$

e) $20.1 \times 10^{-12} Fm^{-1}$

8 The tangent of any point in the electric field lines indicates the …………….

a) current.

b) voltage.

c) emf.

d) resistance.

e) electric field intensity.

9. $1 \text{ Tesla} = ?$

a) 10 Gauss

b) 10^2 Gauss

c) 10^3 Gauss

d) 10^4 Gauss

e) 10^{10} Gauss

10. What is the value of the proportionality constant of Biot-Savart Law?

a) $10^{-7}\ TmA^{-1}$

b) $2\pi \times 10^{-7}\ TmA^{-1}$

c) $4\pi \times 10^{-7}\ TmA^{-1}$

d) $2\pi\ TmA^{-1}$

e) $4\pi\ TmA^{-1}$

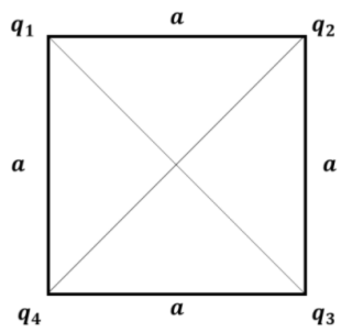
Part 2: SAQ

Time: 1 Hour and 20 Minutes **Total Marks: 20**

(Answer Question No. 1 and any three from the rest.)

[The figures in the right margin indicate full marks for the respective question]

1. See the stem carefully and answer the question that follows:



Assume that,

$q_1 = 1 \times 10^{-8} c$

$q_2 = 2 \times 10^{-8} c$

$q_3 = 2 \times 10^{-8} c$

and $q_4 = 1 \times 10^{-8} c$

also, $a = 10\ cm$

a) Find the potential at center of the following square. [3]

b) Is it possible to find electric field zero in the center of this square? If not, suggest an arrangement to get zero at center. [2]

2. a) What is electrostatics? [1]

b) Explain the term “Quantization of electric charge”. [3]

c) Write down the electric field equation for the system of charge. [1]

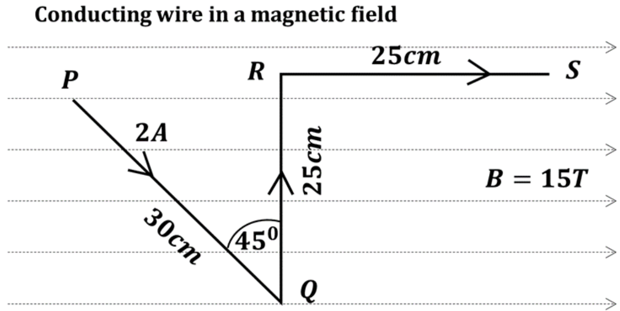
3. a) Define drift velocity of electron. Relate it with current density. [3]

b) Express the Ohm’s law in terms of current density. [2]

4. a) Write down the conclusion of the Oersted’s Experiment. [2]

b) Find the force of each segment of the wire: [3]

Conducting wire in a magnetic field



5. a) State and explain the Ampere’s law. [2]

b) A solenoid is 1.0m long and 3cm in mean diameter. It has five layers of windings of 850 turns each and carries current of 5A. (a) What is B at its centre? (b) What is the magnetic flux φ_B for a cross-section of the solenoid at this centre? [3]